

Padstow Simplified Section Profiler

QGIS cheat sheet for producing the deliverables for each area (Area 1 to Area 8).

Final outputs per area

1. **ortho.jpg** - ortho background
2. **dsm.png** - coloured DSM image
3. **section_lines.png** - transparent overlay containing only section lines and labels
4. **areaX_section_profiles.csv** - sampled profile data for the chart

Golden rule: the three images must all come from the **same extent, same layout map, and same pixel size**. If not, the overlay will drift.

1. Set up the area in QGIS

- Load the ortho, the DSM, and your fixed section line layer.
- You do **not** need contour lines for this simplified app.
- Style the DSM as **Singleband pseudocolor**, choose the colour ramp, set sensible min/max values, and click **Classify**.

2. Create the 3 fixed section lines

- Create a new **LineString** layer. Do not include Z or M values.
- Add these fields: **section_id** (text), **label** (text), **sort_order** (integer).
- Draw the 3 fixed lines for that area. Keep them straight, deliberate, and client-friendly.
- Use consistent IDs, for example Area 8: **A8-01**, **A8-02**, **A8-03**.
- Style the lines in a bold colour so they are easy to export as an overlay.

3. Sample the DSM heights

- Run **Points along geometry** on the section line layer.
- Use a spacing of **1 metre** unless you want a lighter dataset.
- Run **Sample raster values** using the point layer and the DSM raster.
- Open the sampled points attribute table and create clean export fields.

Field	Type	Expression
height_m	Decimal, 2 dp	round("your_sampled_field_name", 2)
distance_m	Whole number or decimal	"distance"

- Export the sampled points layer as CSV and keep these fields: **section_id**, **label**, **sort_order**, **distance_m**, **height_m**.
- Name the CSV like **area8_section_profiles.csv**, **area3_section_profiles.csv**, and so on.

4. Export the three images from one fixed layout

- Use one QGIS Print Layout for that area and keep the map frame fixed once it is set.
- **ortho.jpg**: turn ortho ON, DSM OFF, section lines OFF.
- **dsm.png**: turn ortho OFF, DSM ON, section lines OFF.

- **section_lines.png**: turn ortho OFF, DSM OFF, section lines ON.
- The overlay PNG must have a **transparent background** and contain only the lines and labels you want visible in the app.

5. Make sure the section overlay is actually transparent

- In the layout, click the map item and untick **Background**.
- Also check the page itself is not forcing a white background.
- Turn off any legends, white rectangles, or extra items when exporting the line-only PNG.
- Open **section_lines.png** afterwards. If the background is white, it is wrong. If it is transparent, it is right.

6. Check alignment before uploading

- Check the pixel dimensions of **ortho.jpg**, **dsm.png**, and **section_lines.png**. They should all match.
- If one image is a different size, the overlay will drift.
- If the sizes match but the lines are still off, re-export the images from the same exact extent again.

7. Put the files in the correct folder

Folder structure per area
assets/area8/ortho.jpg
assets/area8/dsm.png
assets/area8/section_lines.png
assets/area8/area8_section_profiles.csv

Quick repeatable workflow

- Load ortho and DSM.
- Style the DSM.
- Create the 3 fixed section lines.
- Run Points along geometry.
- Run Sample raster values.
- Create height_m and distance_m.
- Export areaX_section_profiles.csv.
- Export ortho.jpg, dsm.png, and transparent section_lines.png from the same extent.
- Check all 3 image sizes match.
- Place the files into the correct assets/areaX folder.

Common mistakes to avoid

- Exporting the ortho, DSM, and overlay from slightly different extents.
- Using a section overlay PNG with a white background instead of a transparent one.
- Leaving CSV field names messy or inconsistent.
- Using image files with different pixel sizes.
- Using section IDs in the CSV that do not match the app config.

Before you call an area finished
Have I exported ortho.jpg ?

Have I exported **dsm.png**?

Have I exported **transparent section_lines.png**?

Have I exported **areaX_section_profiles.csv**?

Do all three images line up and share the same pixel size?

Do the section IDs in the CSV match the app config?